

Volume IV · Chapter 18 — Ingestion, OCR & Document Understanding · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-IV-Chapter-18-context.md for the AI interaction log.

Prerequisites: Ch 5, Ch 12.

Learning outcomes — the student can: parse born-digital and scanned medical books; run OCR handling medical vocabulary; extract layout/structure (sections, tables, references); build a clean, provenance-tagged text corpus.

Labs (hands-on) — 14 h:

#	Lab	h
18a	Extract text from a born-digital medical PDF (PyMuPDF)	3
18b	OCR a scanned page; measure & clean errors (Tesseract / doc model)	4
18c	Layout-aware parsing (unstructured/Nougat/Marker) into sections	4
18d	Build a provenance-tagged text corpus (page/section metadata)	3

Datasets/tools: licensed/open medical texts; PyMuPDF, Tesseract, unstructured/Nougat/Marker. **Assessment:** ingestion pipeline + corpus (60%); OCR-quality report (20%); quiz (20%). **Key decisions:** born-digital vs. scanned handling; OCR engine; how much structure to preserve; **licensing (owned/licensed sources only)**. **References:** plan §6; §13. **Hours:** Theory 8 + Lab 14 = 22.